The Sticky Kakeya Conjecture

S.N. Bhatt Presentation (ICTS)

Saptarshi Mandal

Indian Institute of Technology, Kanpur

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Mentor: Trishen Gunaratnam

Kakeya Set

Definition (Kakeya Set)

$K \subseteq \mathbb{R}^n$ that contains a unit line segment in every direction.

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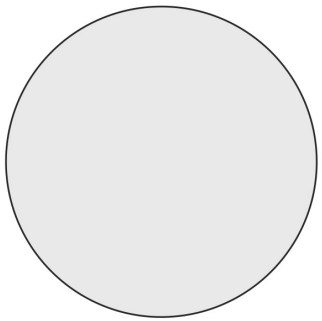
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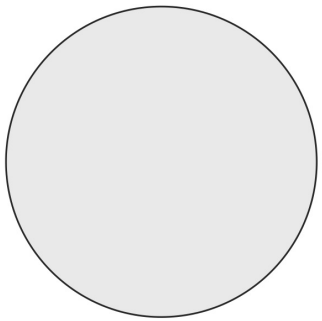


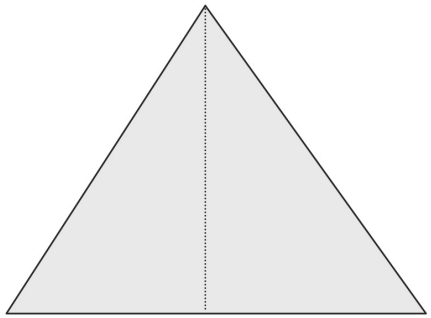
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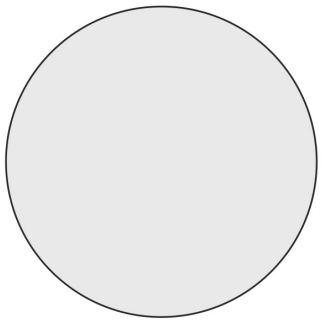


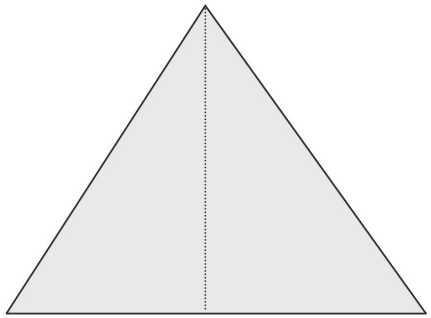
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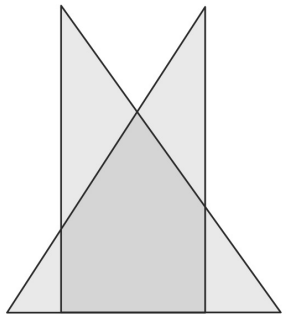
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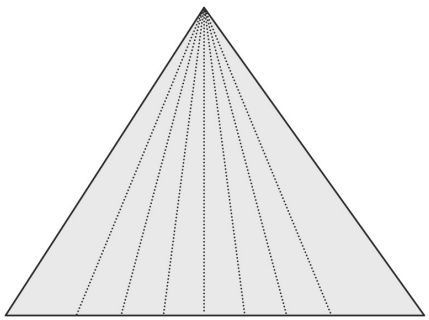


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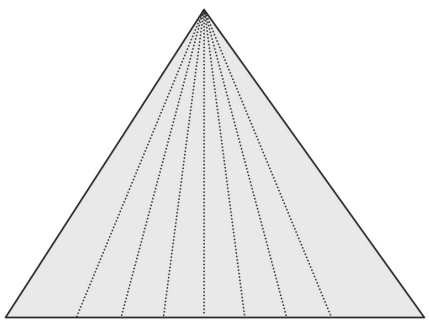


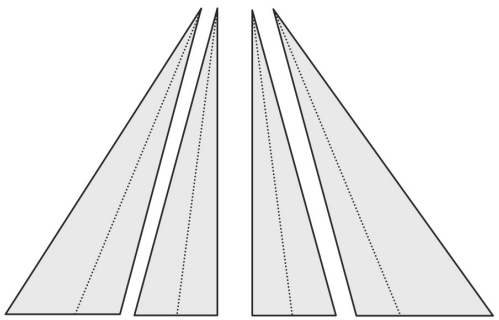
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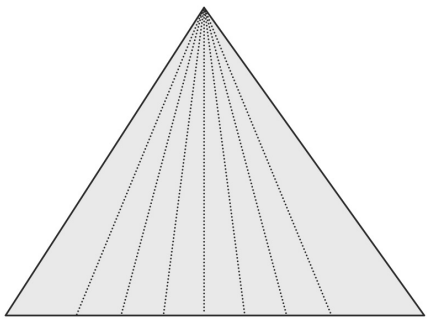


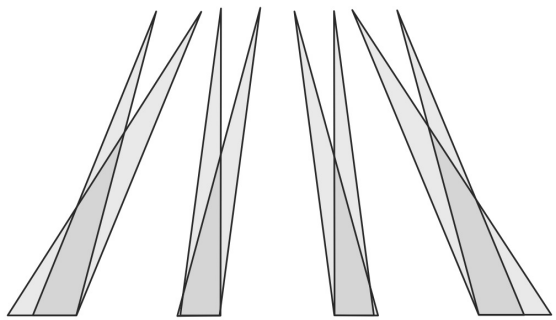
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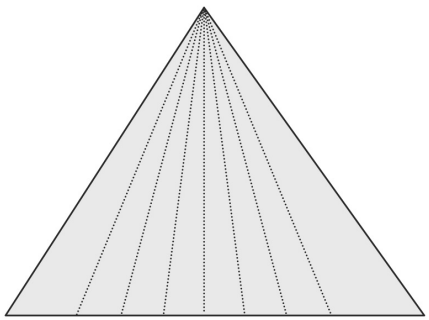


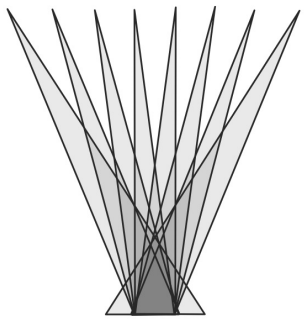
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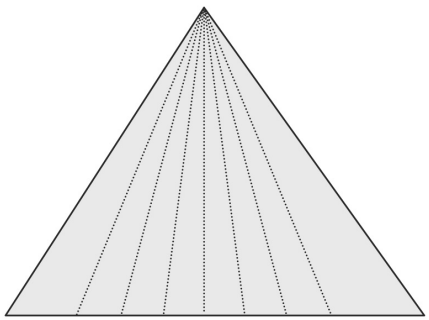


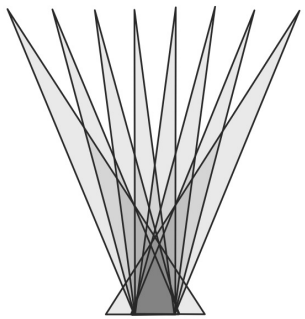
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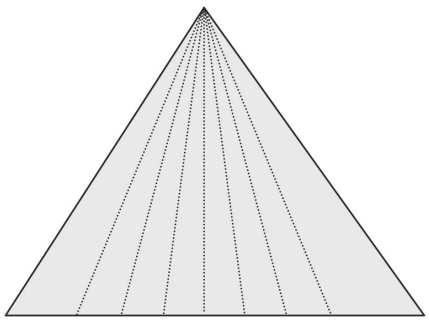
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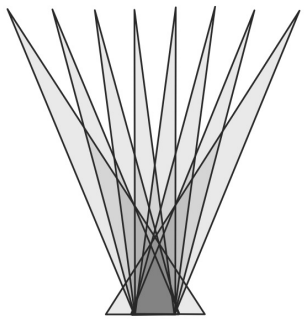
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Theorem (Besicovitch)

There are Kakeya Sets of measure zero.

So, area/volume cannot capture the "bigness" of Kakeya Sets.

Hausdorff Dimension



$$K = \bigcap K_n$$

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Conjecture (Kakeya)

If $K \subseteq \mathbb{R}^n$ is a Kakeya set, then $\dim_{\mathcal{H}}(K) = n$.

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Conjecture (Kakeya), $n = 3$

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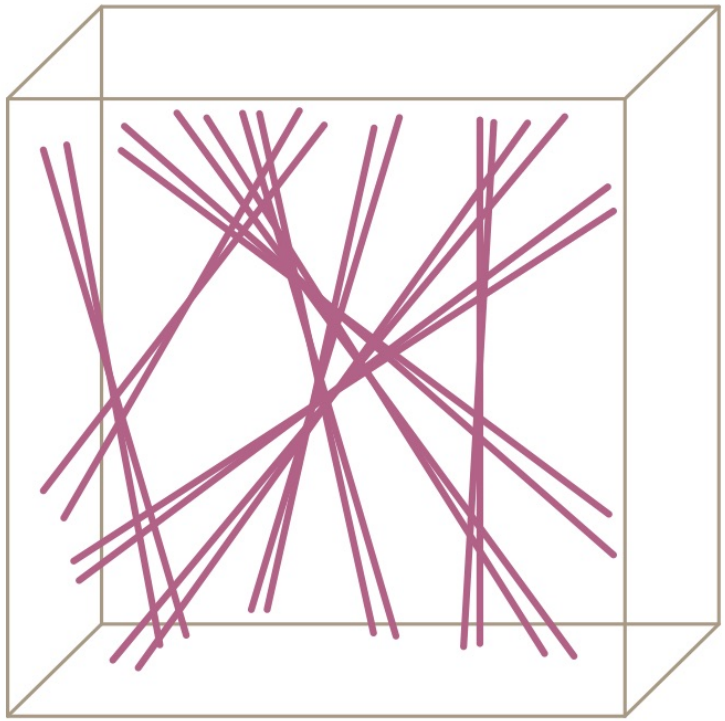
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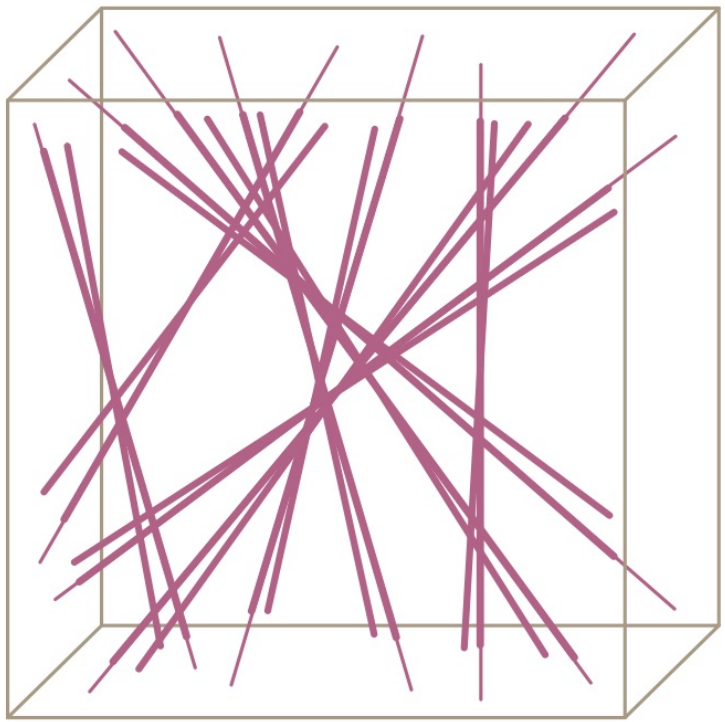
Sticky Kakeya Sets

We will look at a scales $\delta > 0$ inside $[-1, 1]^3$.



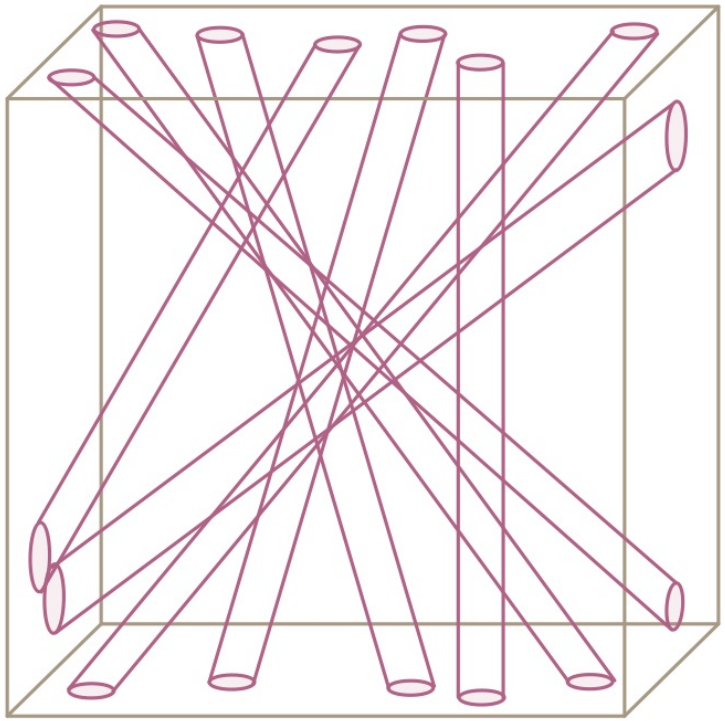
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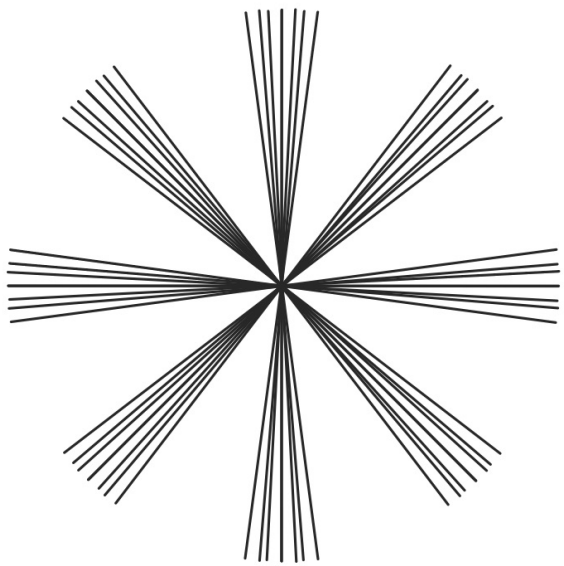
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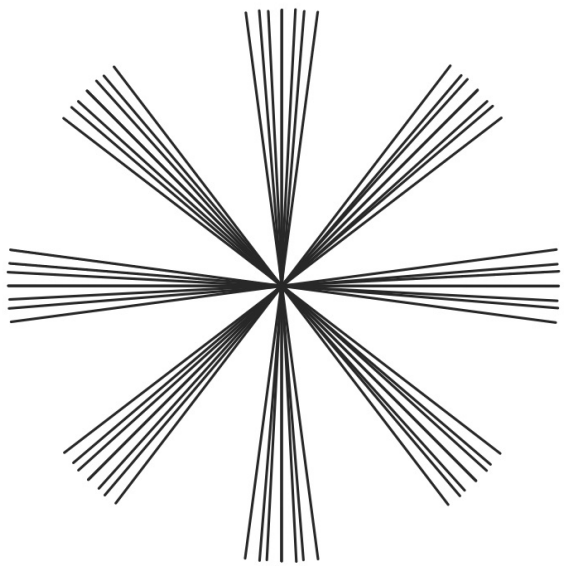
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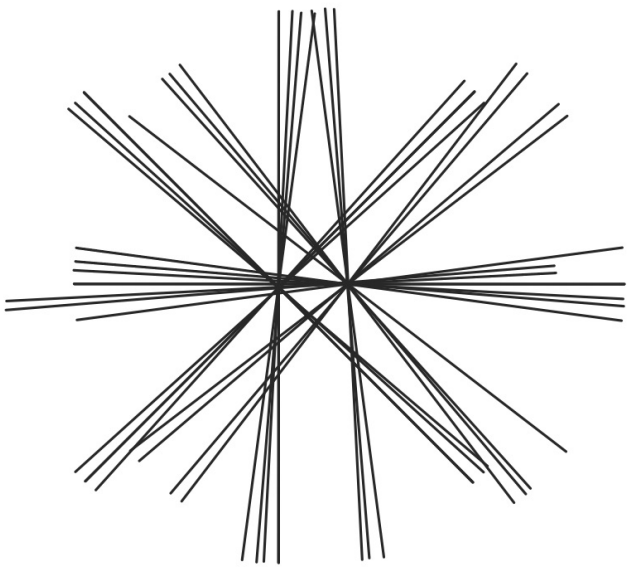


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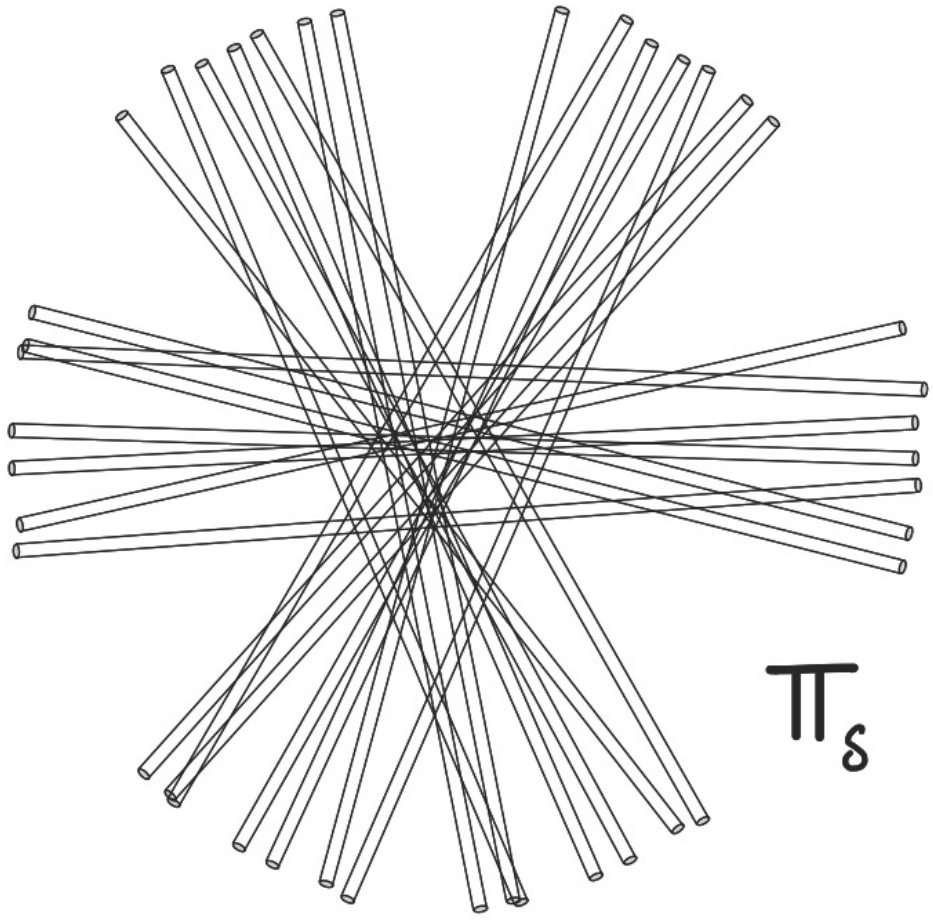
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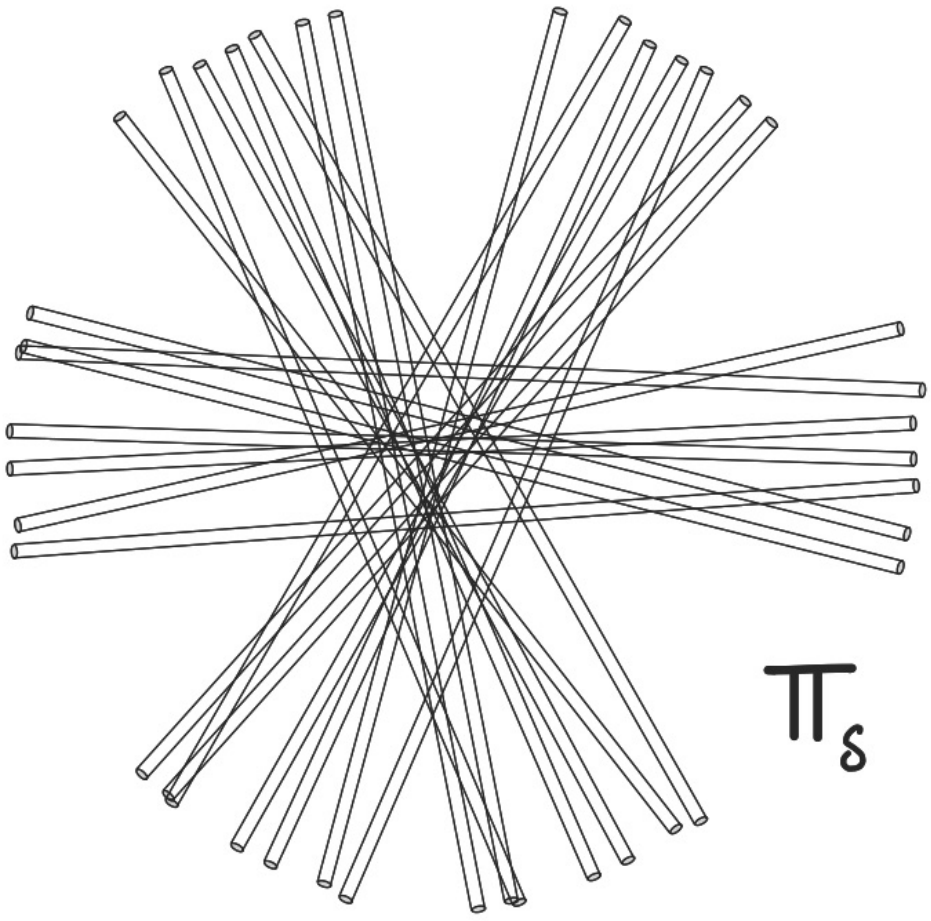
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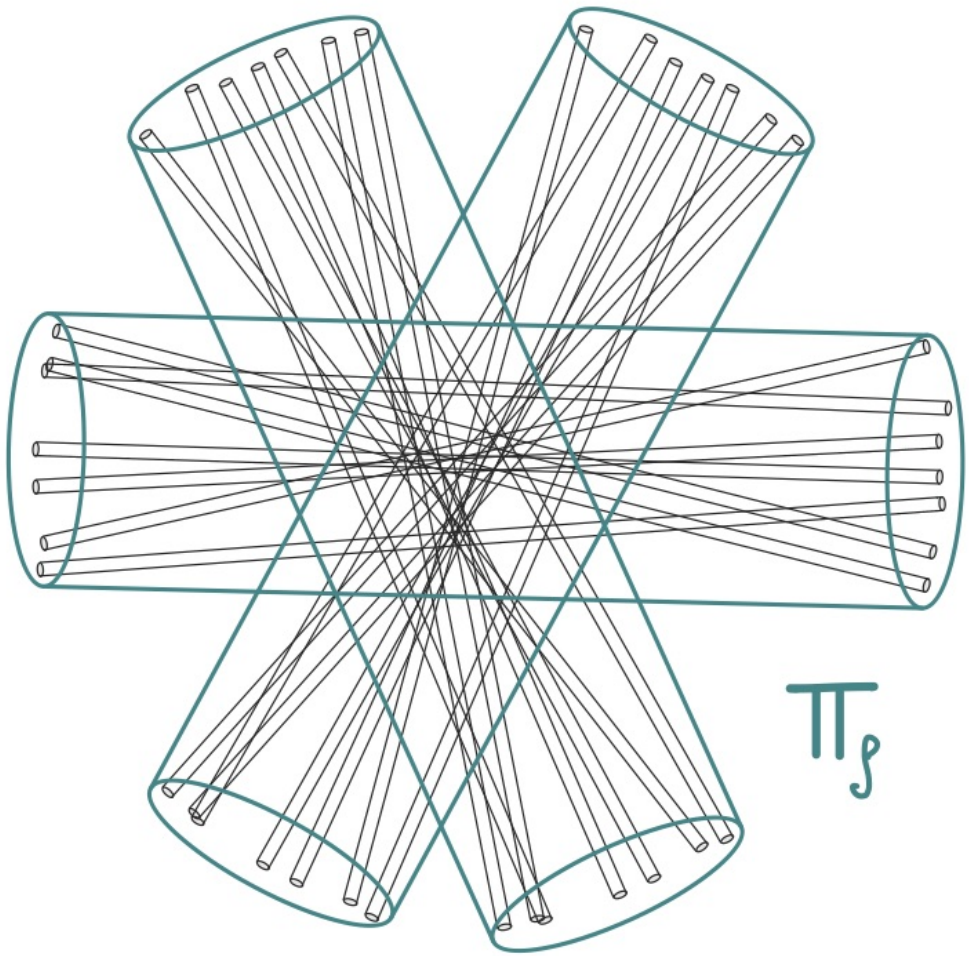
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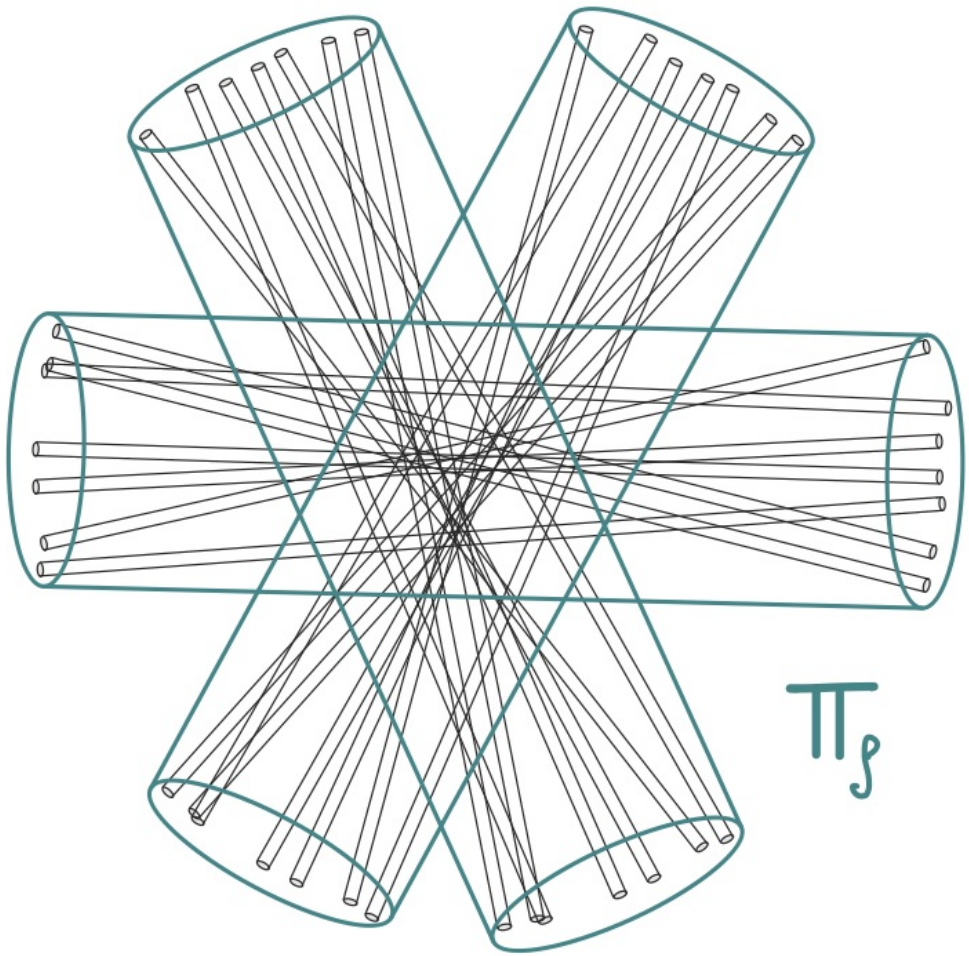
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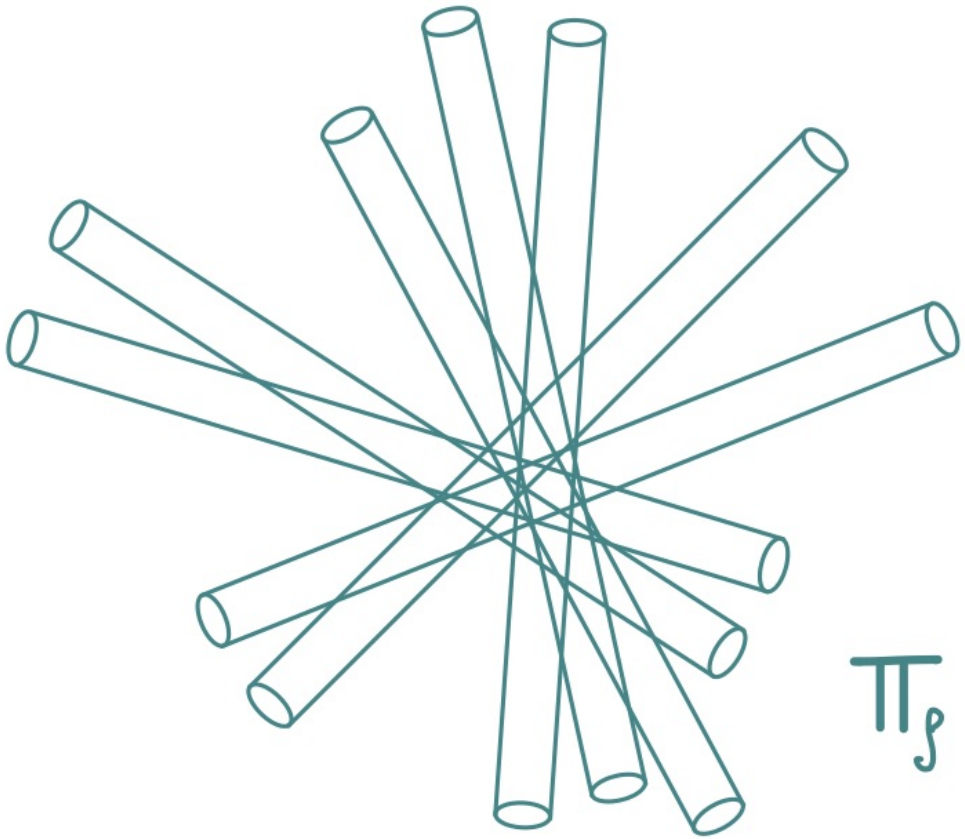
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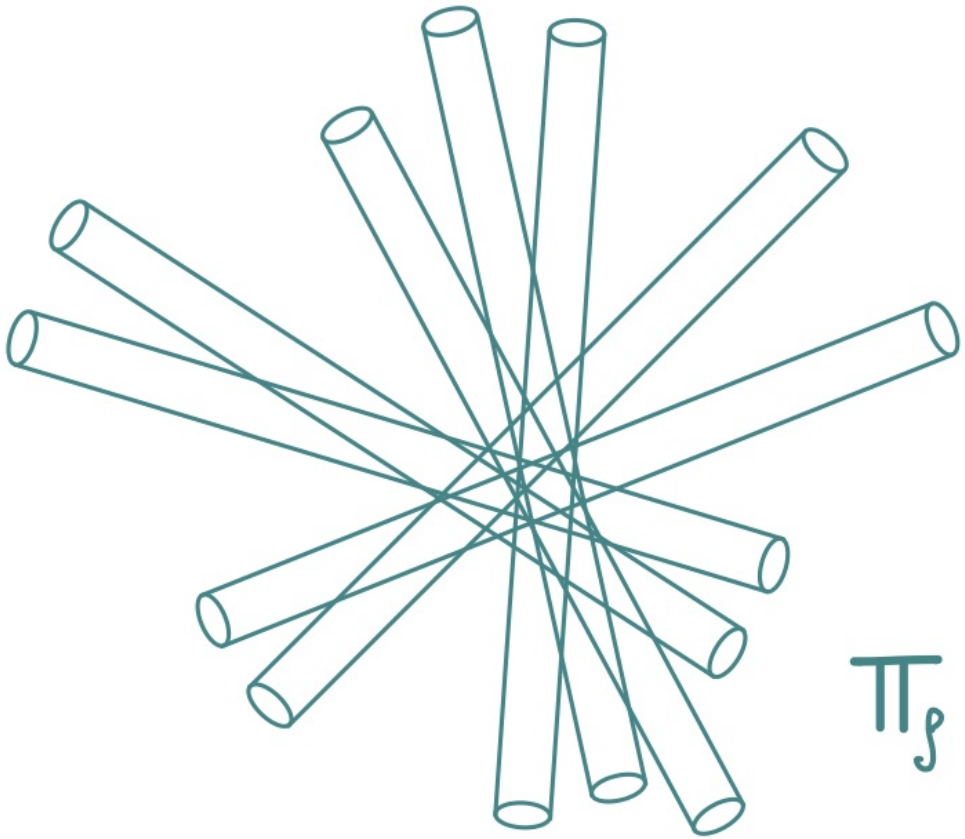
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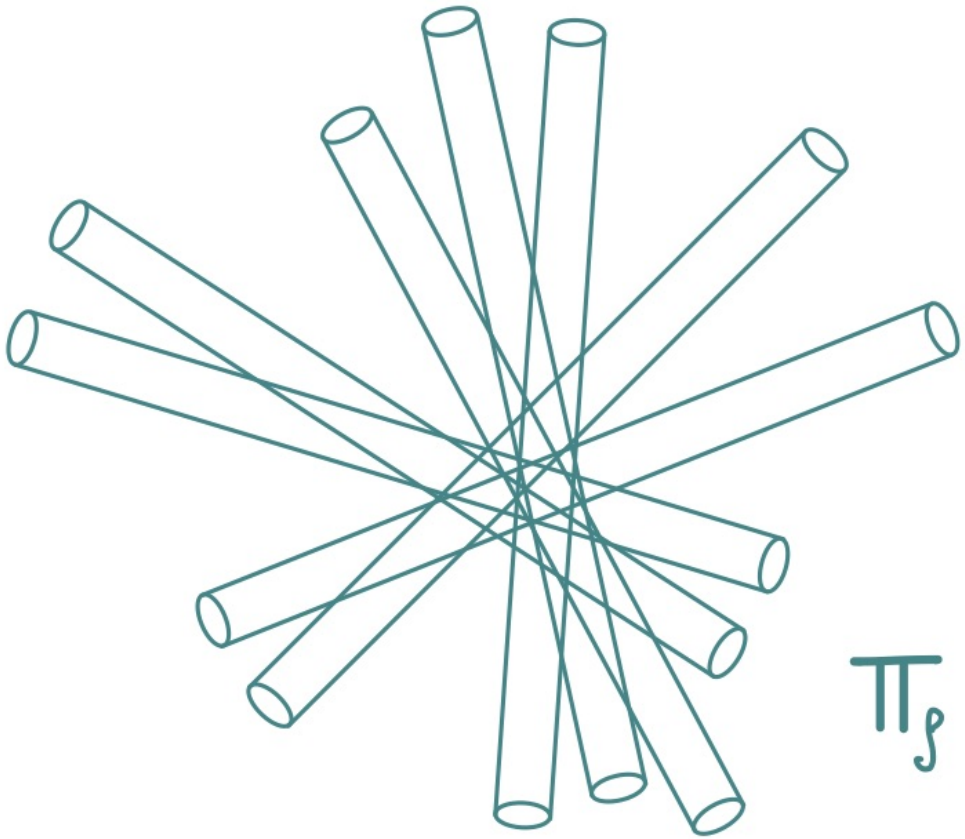


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$$\left|\bigcup \mathbb{T}_\rho\right| \sim \rho^\sigma$$

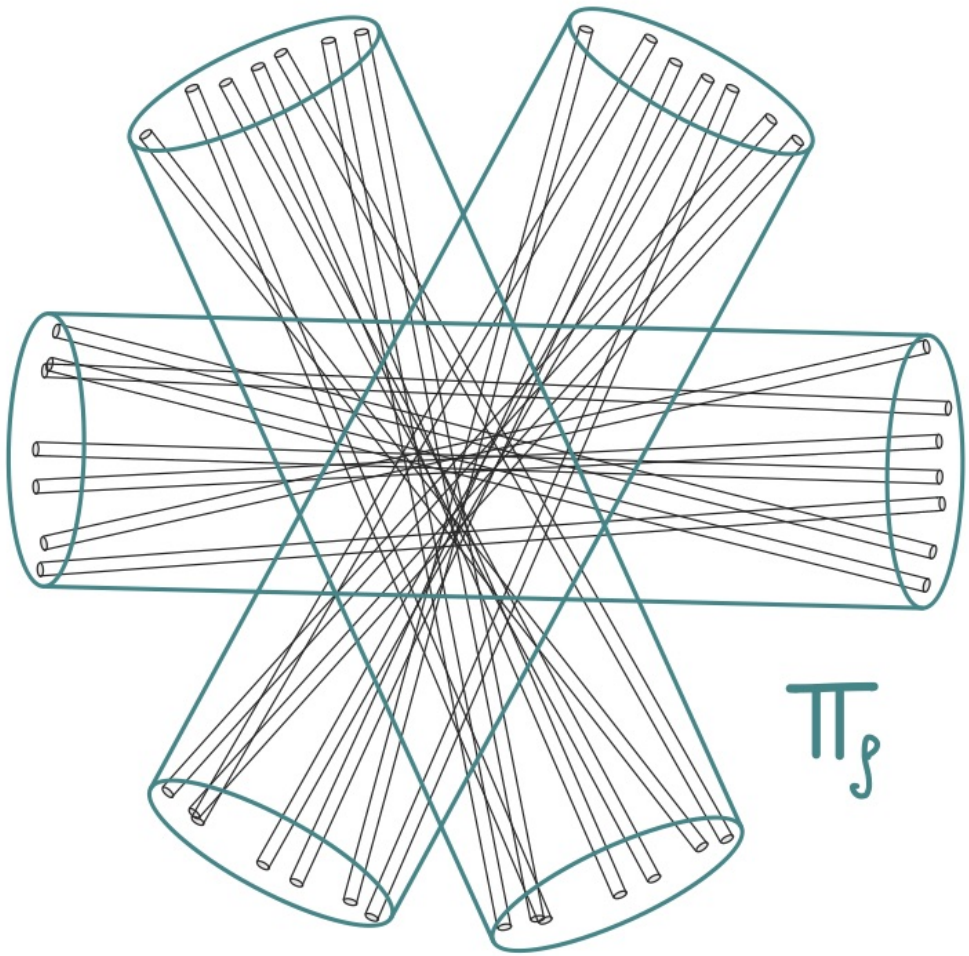


Coarse Self-Similarity ($\Leftarrow$ Stickiness)

$\mathbb{T}_\delta$ be sticky and attain σ. Then $\forall \rho > \delta$, $\exists\ \mathbb{T}_\rho$ such that:

1. $\mathbb{T}_\rho$ covers $\mathbb{T}_\delta$
2. $\mathbb{T}_\rho$ is sticky
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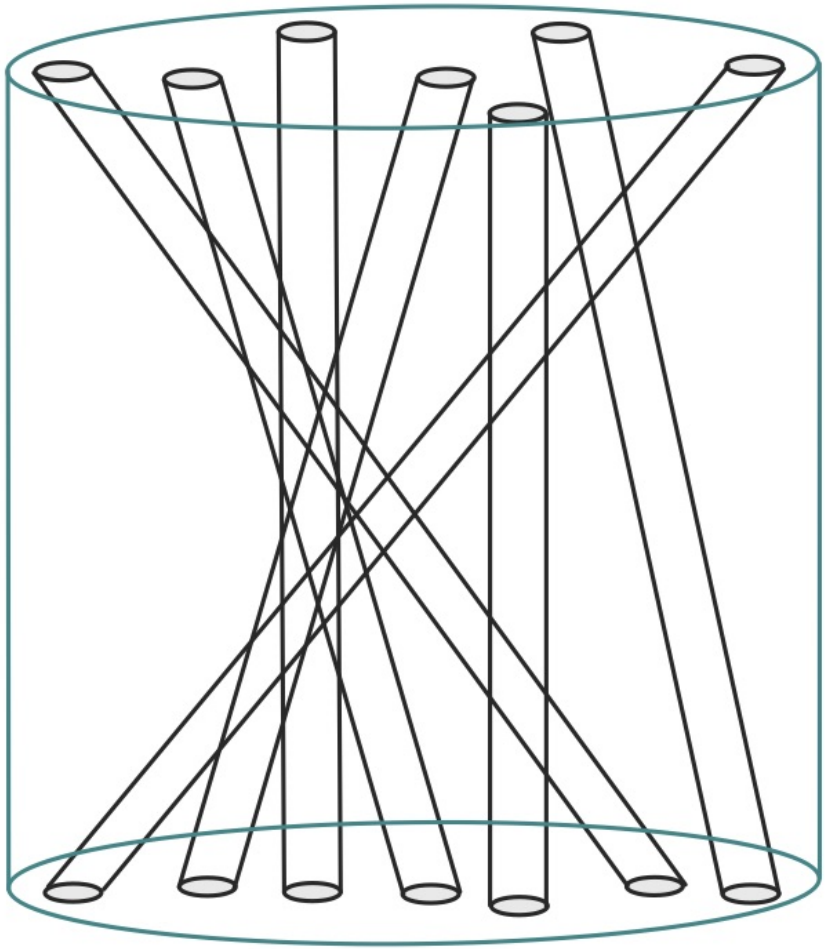


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$\pi \frac{\delta}{\rho}$

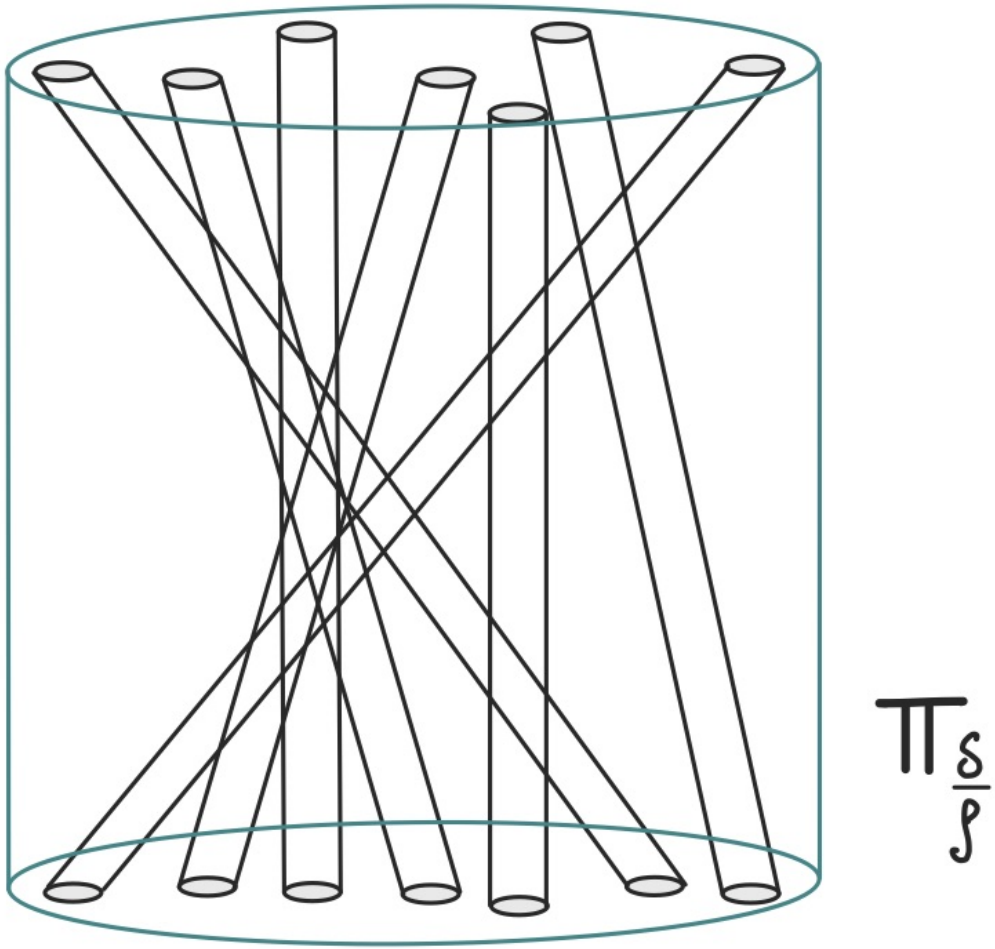
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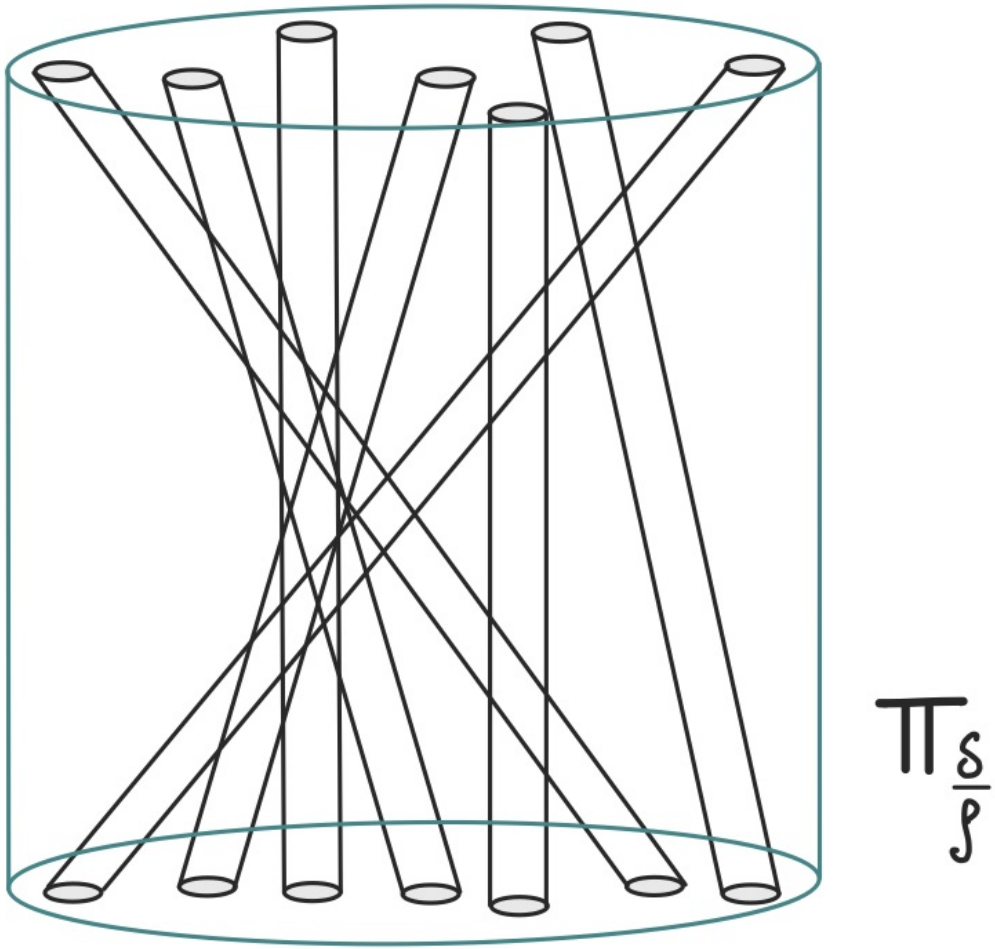
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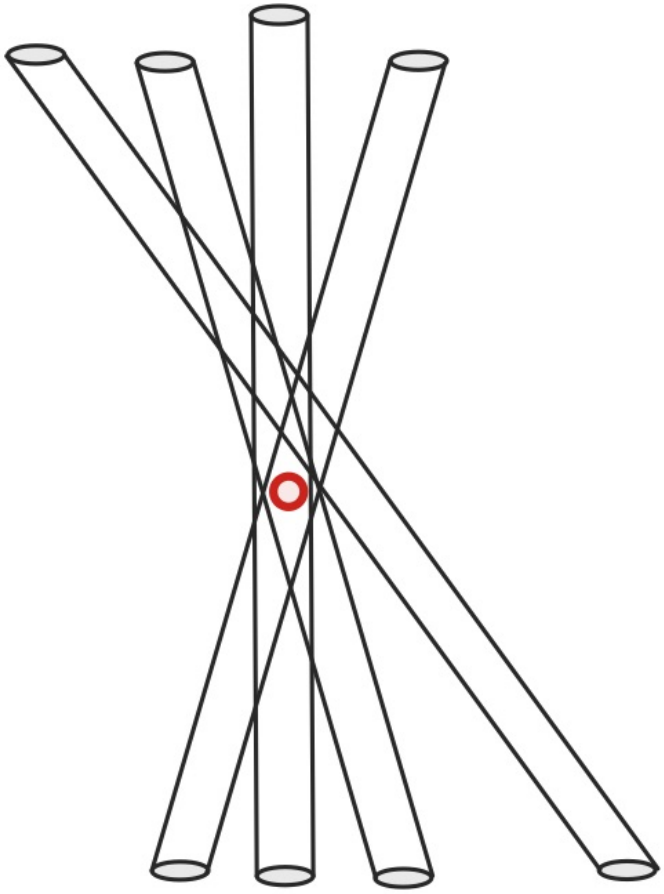
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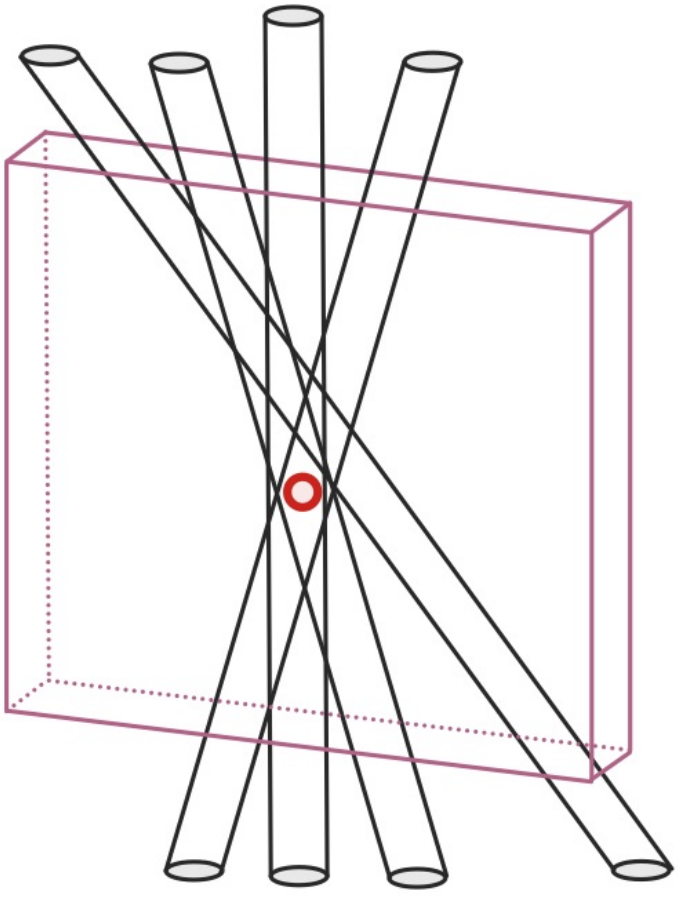
$$\left|\bigcup \mathbb{T}_{\delta/\rho}\right| \sim \left(\frac{\delta}{\rho}\right)^\sigma$$



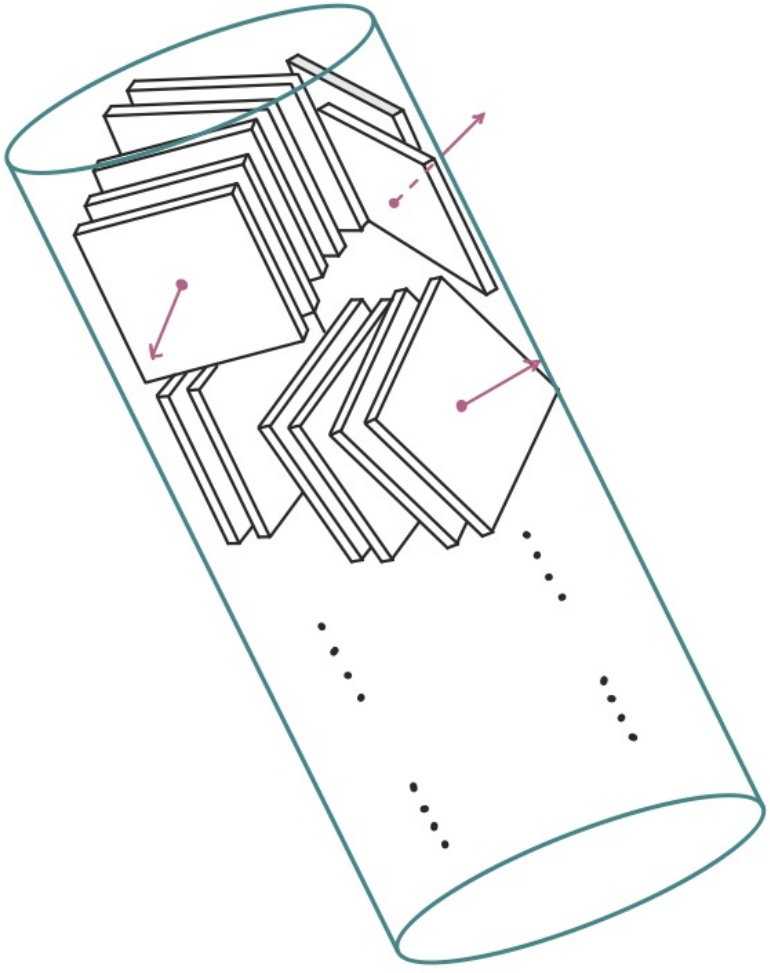
Plaininess



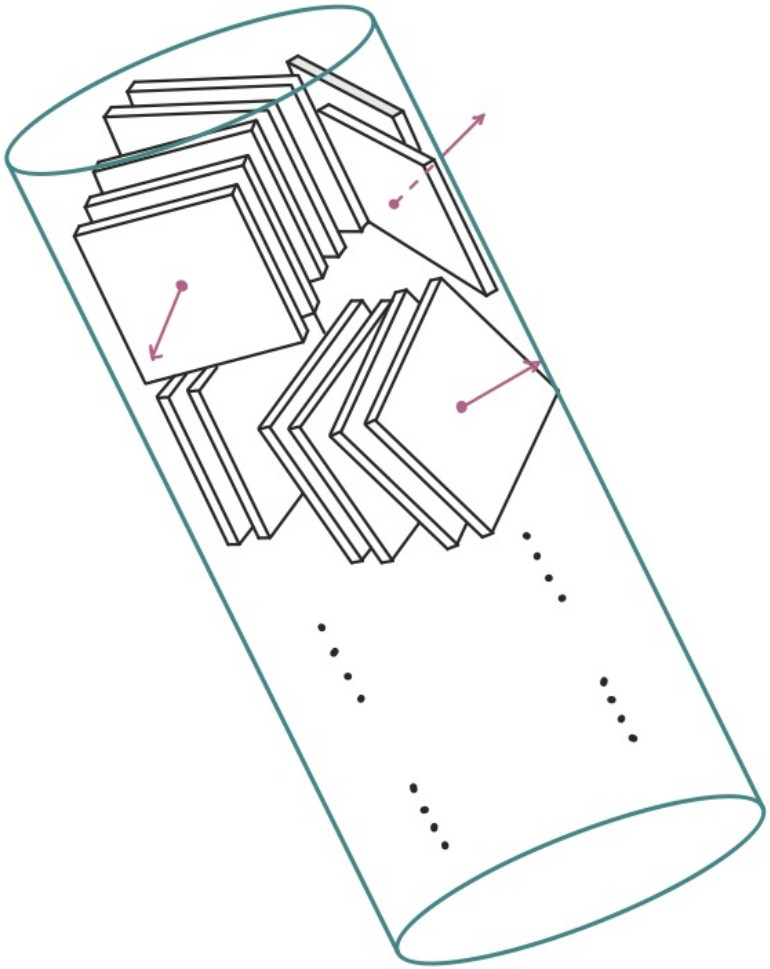
Plaininess



Plaininess

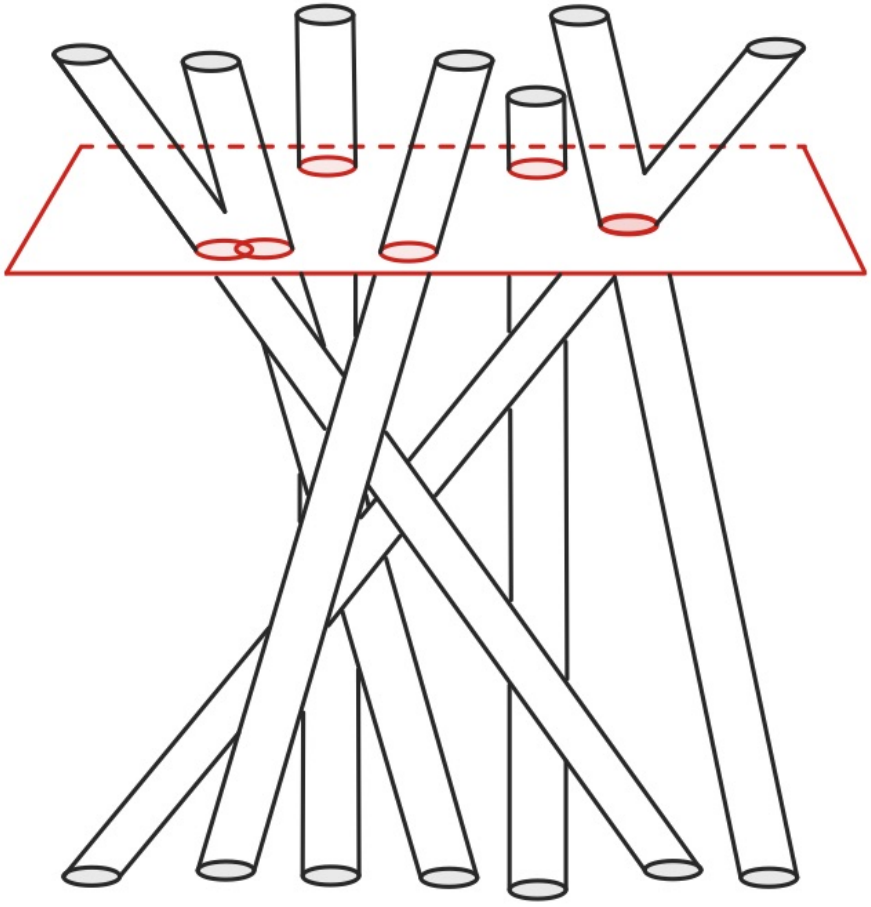


Plaininess

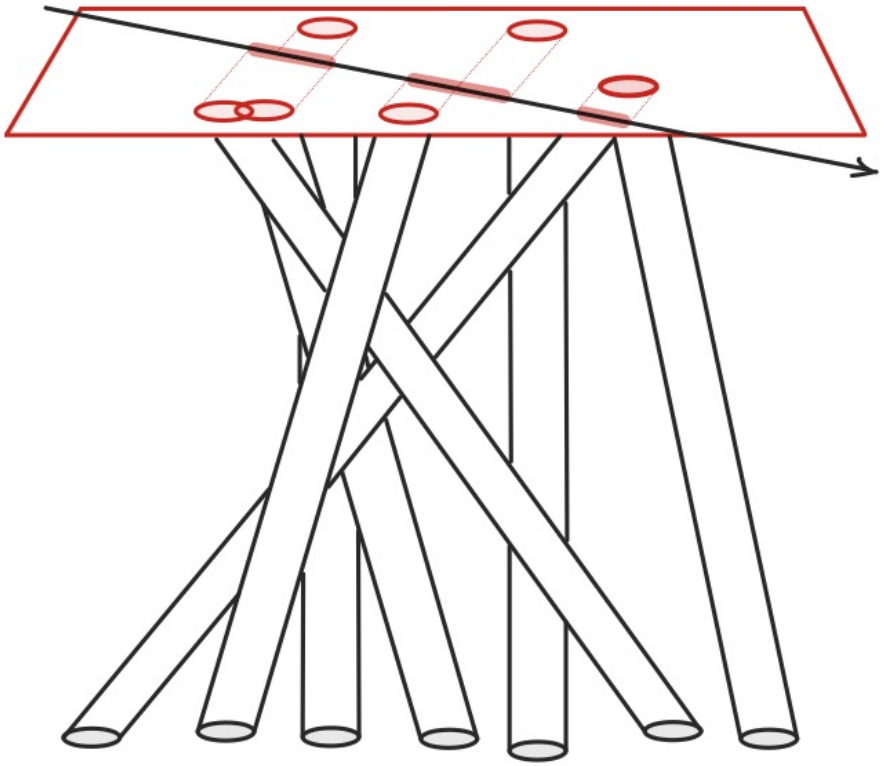


$\longrightarrow$ changes in a Lipschitz manner with the point.

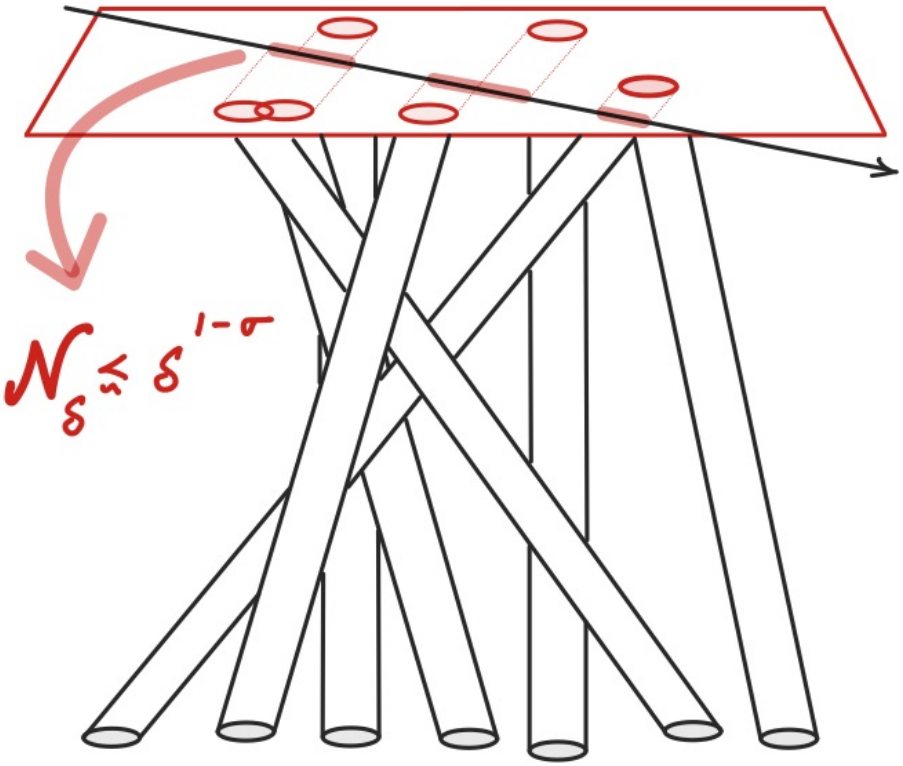
Graininess



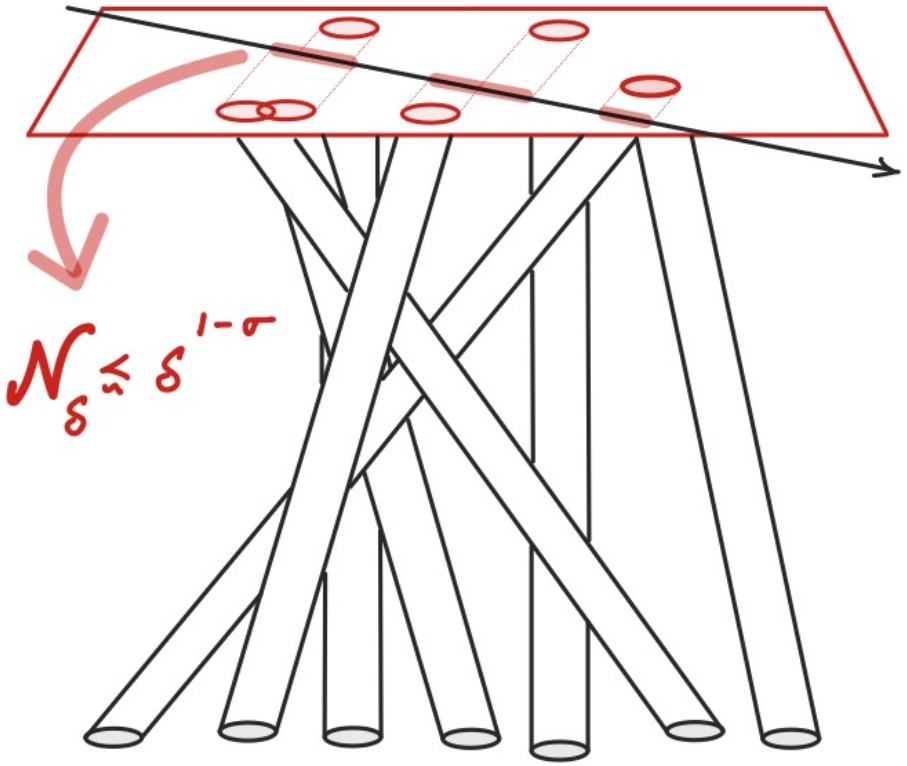
Graininess



Graininess

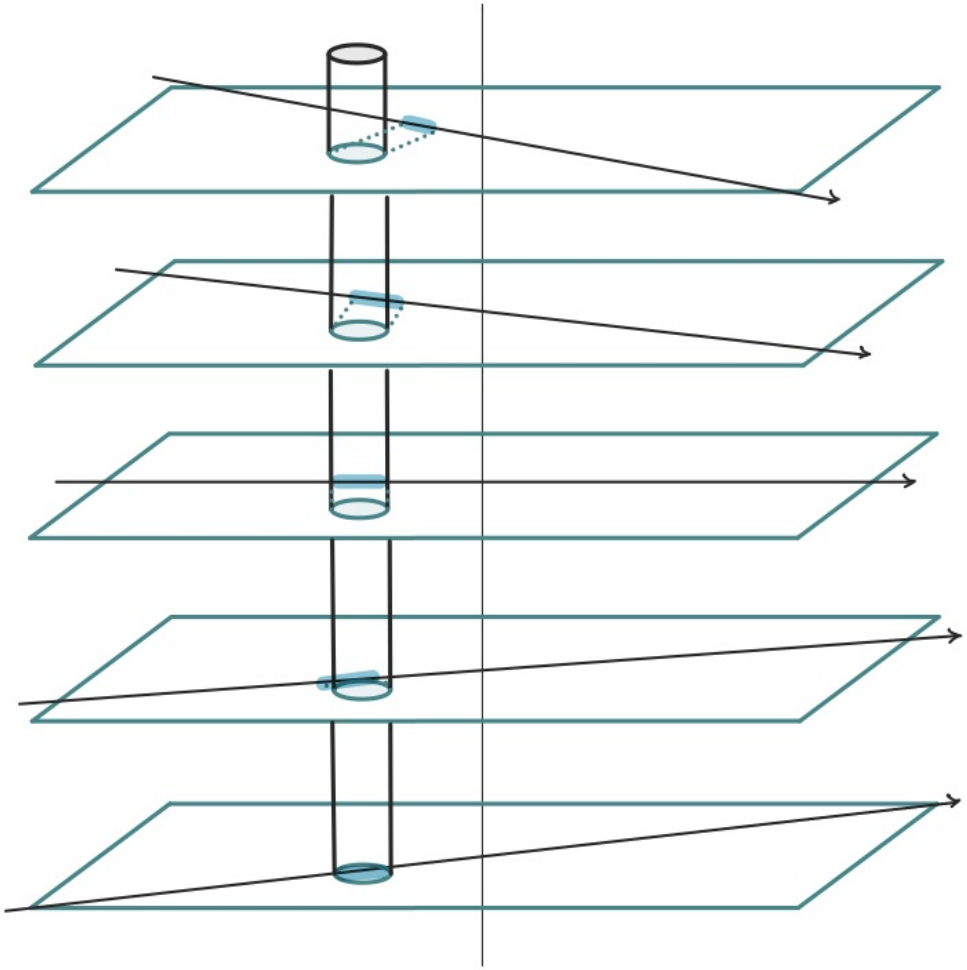


Graininess

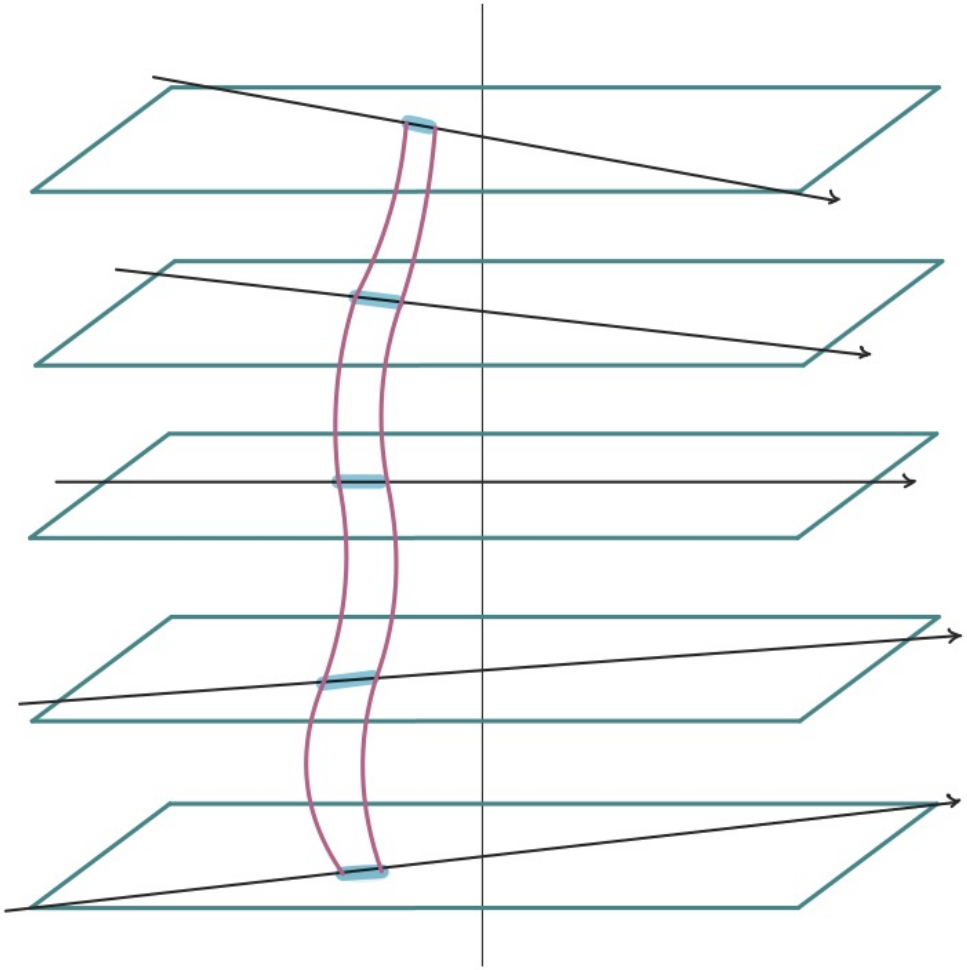


$\longrightarrow$ changes in a C^2-manner with height.

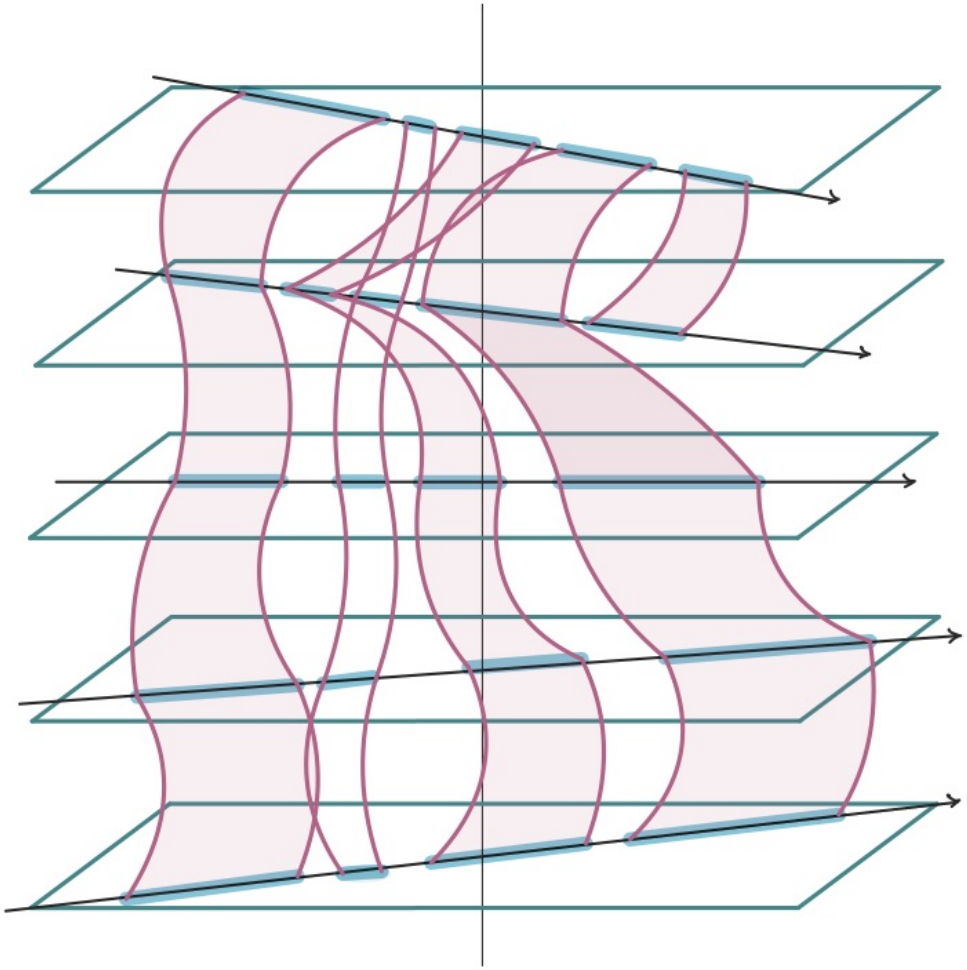
Contradicting Wolff



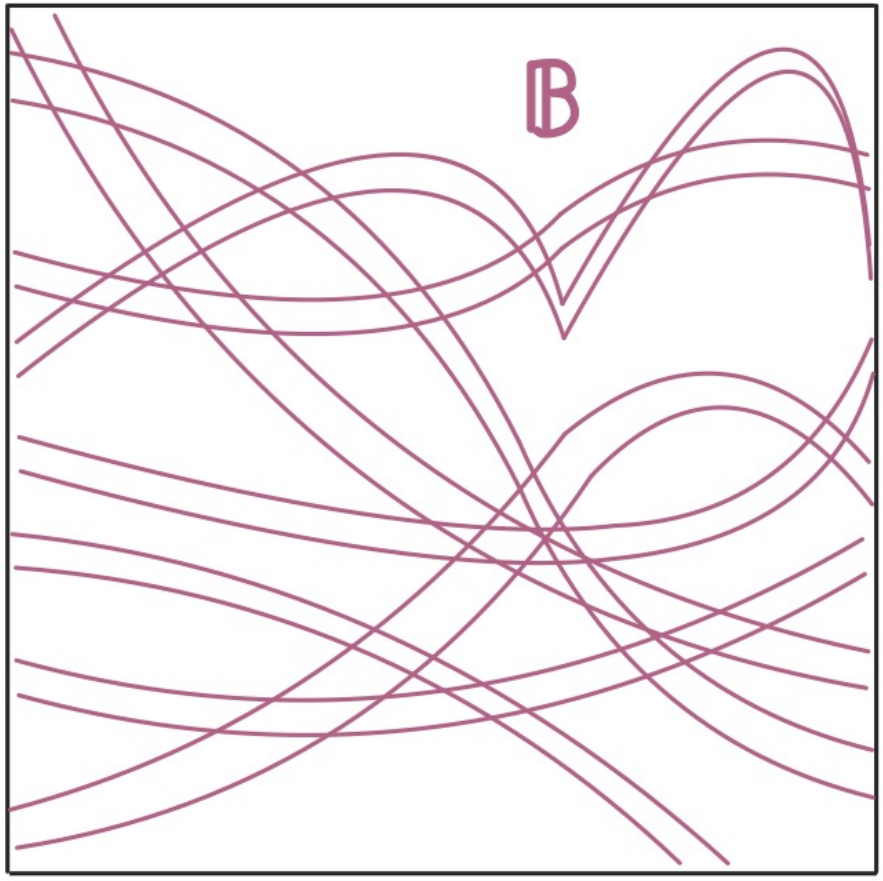
Contradicting Wolff



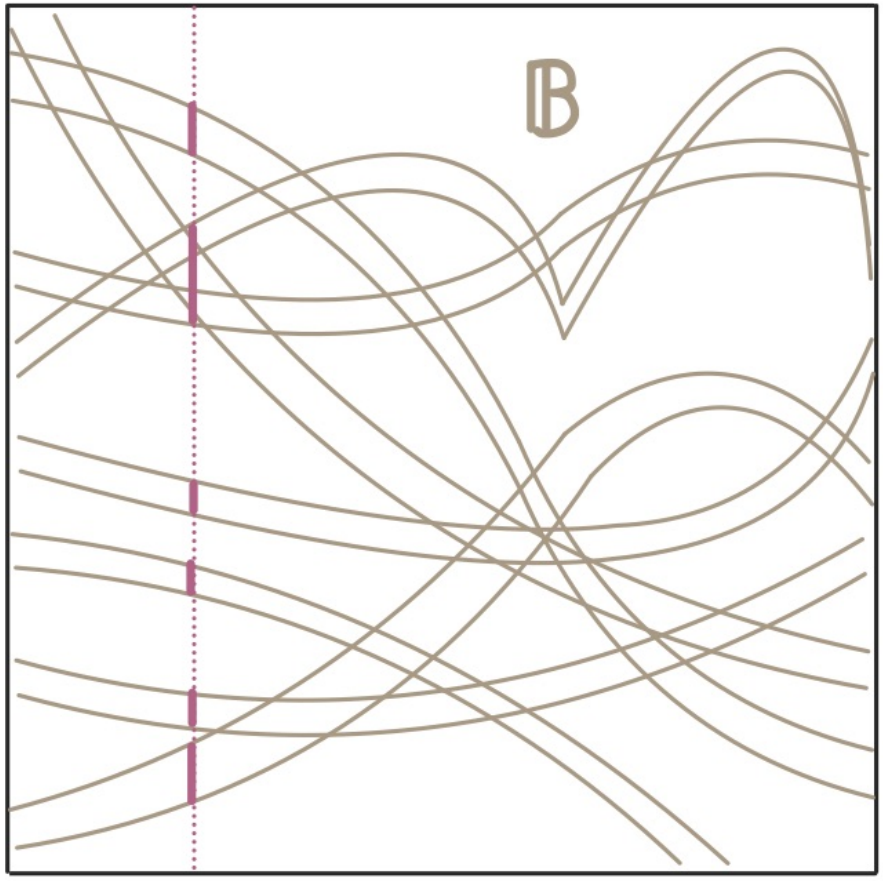
Contradicting Wolff



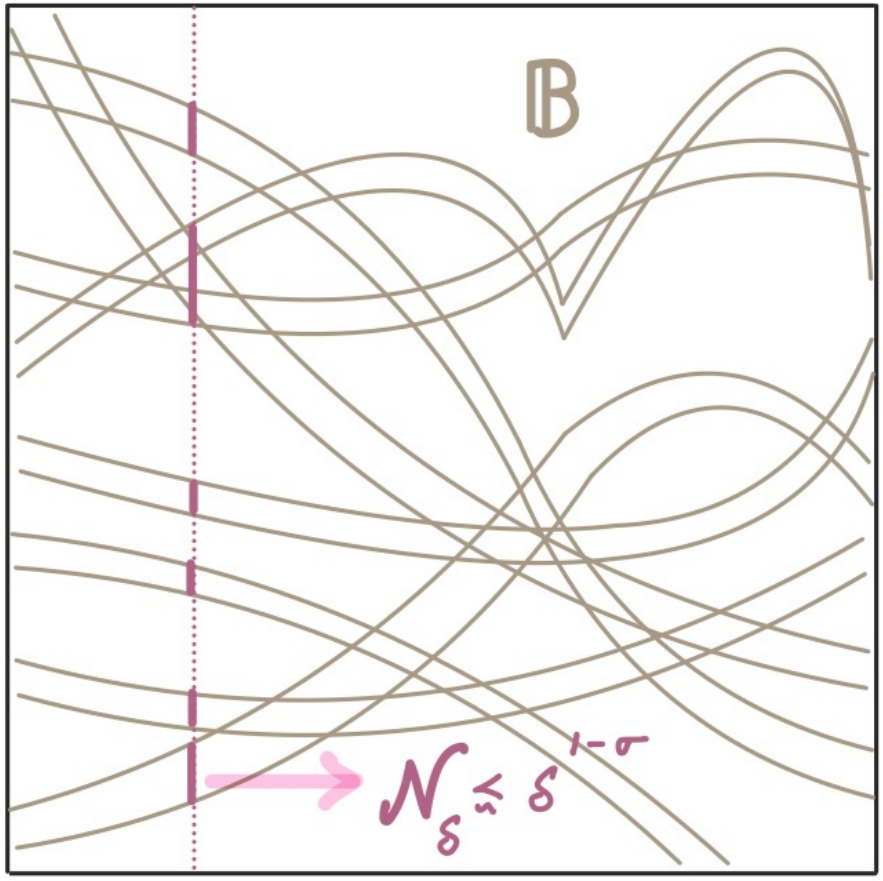
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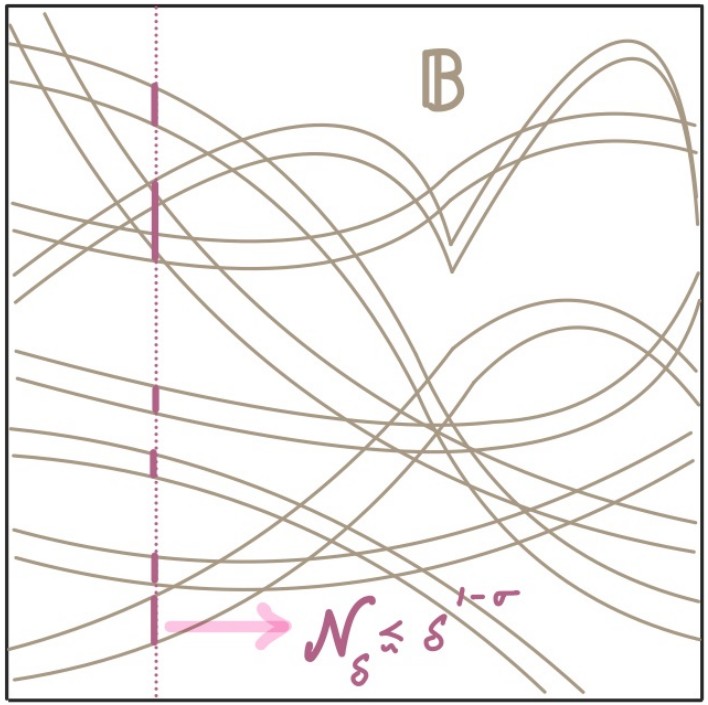
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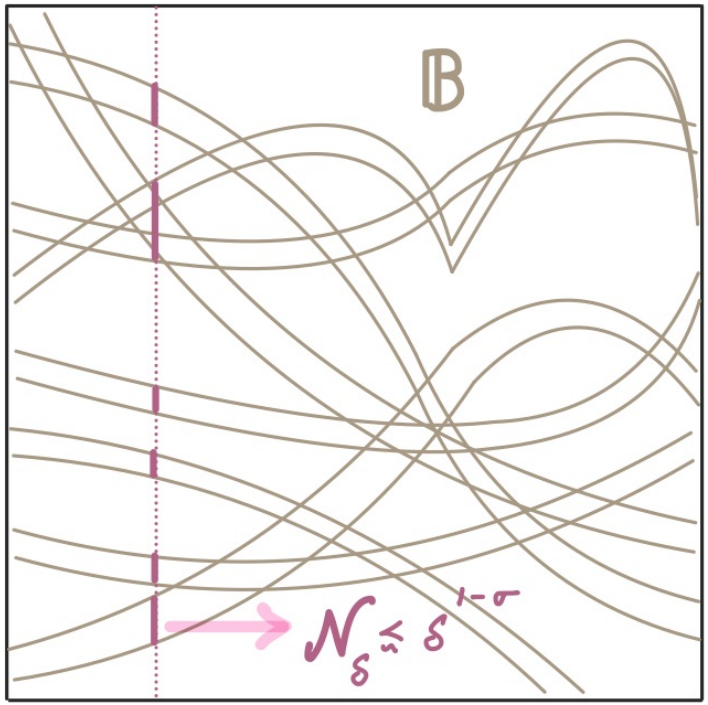


Contradicting Wolff



1. $\mu :=$ avg multiplicity

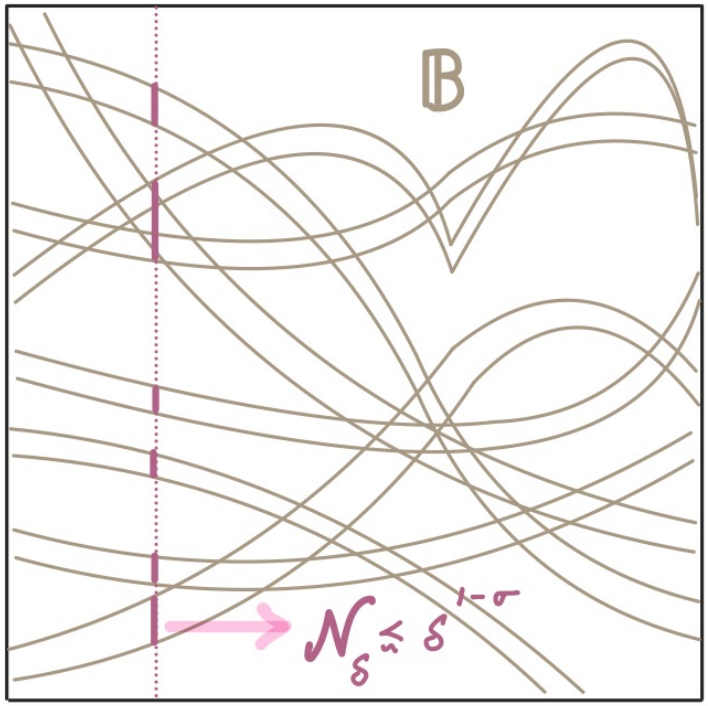
Contradicting Wolff



1. $\mu :=$ avg multiplicity
2. Coarse self-similarity forces

$$\delta^{-1-\sigma} \leq \mu$$

Contradicting Wolff



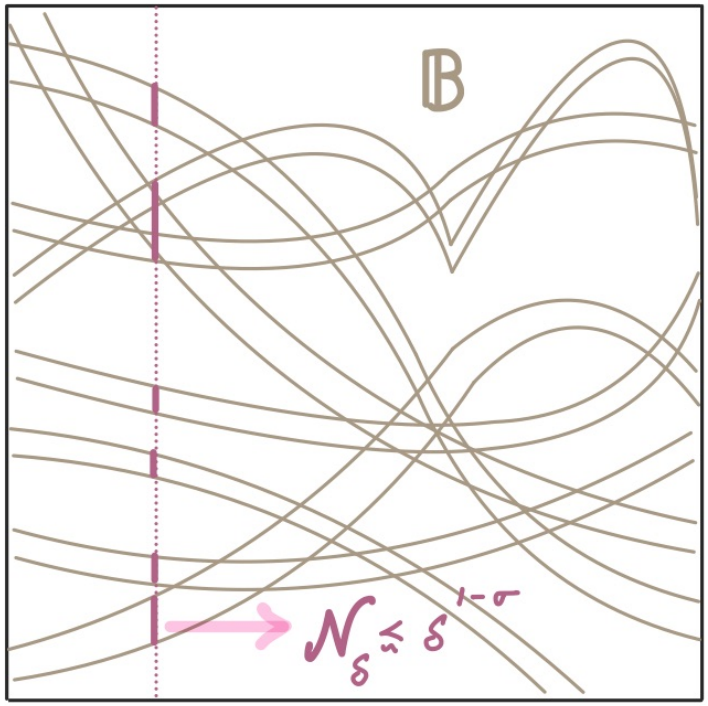
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Contradicting Wolff



1. $\mu :=$ avg multiplicity
2. Coarse self-similarity forces
$$\delta^{-1-\sigma} \leq \mu$$
3. By Wolff's bound (for 2D Kakeya)
$$\mu \leq \delta^{-1}$$

Since this holds for all sufficiently small δ, we have $\sigma = 0$. $\square$

Thank You!

questions?



Appendix A: The Proof

$$\delta^{-1} \leq \left\| \sum_{B \in \mathbb{B}} \chi_B \right\|_{L^1(\bigcup \mathbb{B})}$$

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$$\delta^{-1} \leq \left\| \sum_{B \in \mathbb{B}} \chi_B \right\|_{L^1(\bigcup \mathbb{B})} \leq \left| \bigcup \mathbb{B} \right|^{1/3} \cdot \left\| \sum_{B \in \mathbb{B}} \chi_B \right\|_{L^{3/2}}$$

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So, $\sigma = 0$. $\square$

Appendix B: Why should I care?

Kakeya conjecture is related closely to Fourier analysis.

1. Fefferman used a variant of the Perron tree to show

$$\widehat{Tf} = \chi_B \hat{f}$$

   the ball multiplier is unbounded on $L^p(\mathbb{R}^2)$ when $p \neq 2$.
2. Kakeya-type estimated were used by Bourgain to get new bounds for Stein's Fourier restriction problem in $\mathbb{R}^3$.
3. Methods developed in this context were also applied to the Bochner-Riesz Conjecture.
4. Polynomial-method techniques (which is popular in general) was used by Dvir to prove Wolff's finite field Kakeya conjecture.

Appendix C: History

- 1917 Kakeya
- 1971 Davies
- 1995 Wolff
- 2014 Katz, Laba & Tao
- 2016 Guth
- 2022 Wang & Zahl (Sticky)
- 2025 Wang & Zahl

Appendix D: The Heisenberg Group

$$\mathbb{H} = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : \Im(z_3) = \Im(z_1\overline{z_2})\}$$

is a counterexample to the Kakeya Conjecture in $\mathbb{C}^3$.

Surprisingly, it is sticky, plainy as well as grainy. Yet the arguments of the proof cannot be used here since the slice function is not C^2.

The distinguishing feature is that $\mathbb{R}$ cannot be written as a degree 2 extension of a proper subfield.